26.6: Our understanding of the tree of life continues to change based on new data

The discovery that the glass lizard in Figure 26.1 evolved from a different lineage of legless lizards than did snakes is one example of how our understanding of life's diversity is informed by systematics. Indeed, in recent decades, systematists have gained insight into even the very deepest branches of the tree of life by analyzing DNA sequence data.

From Two Kingdoms to Three Domains

Biologists once classified all known species into two kingdoms: plants and animals. Classification schemes with more than two kingdoms gained broad acceptance in the late 1960s, when many biologists recognized five kingdoms: Monera (prokaryotes), Protista (a diverse kingdom consisting mostly of unicellular organisms), Plantae, Fungi, and Animalia. This system highlighted the two fundamentally different types of cells, prokaryotic and eukaryotic, and set the prokaryotes apart from all eukaryotes by placing them in their own kingdom, Monera.

However, phylogenies based on genetic data soon revealed a problem with this system: Some prokaryotes differ as much from each other as they do from eukaryotes. Such difficulties led biologists to adopt a three-domain system. The three domains—Bacteria, Archaea, and Eukarya—are a classification level higher than the kingdom level. The validity of these domains has been supported by many studies, including a recent study that analyzed nearly 100 completely sequenced genomes.

The domain Bacteria contains most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya consists of all the organisms that have cells containing true nuclei. This domain includes many groups of single-celled organisms as well as multicellular plants, fungi, and animals. **Figure 26.21** represents one possible phylogenetic tree for the three domains and some of the many lineages they encompass.

Figure 26.21

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The three domains of life.
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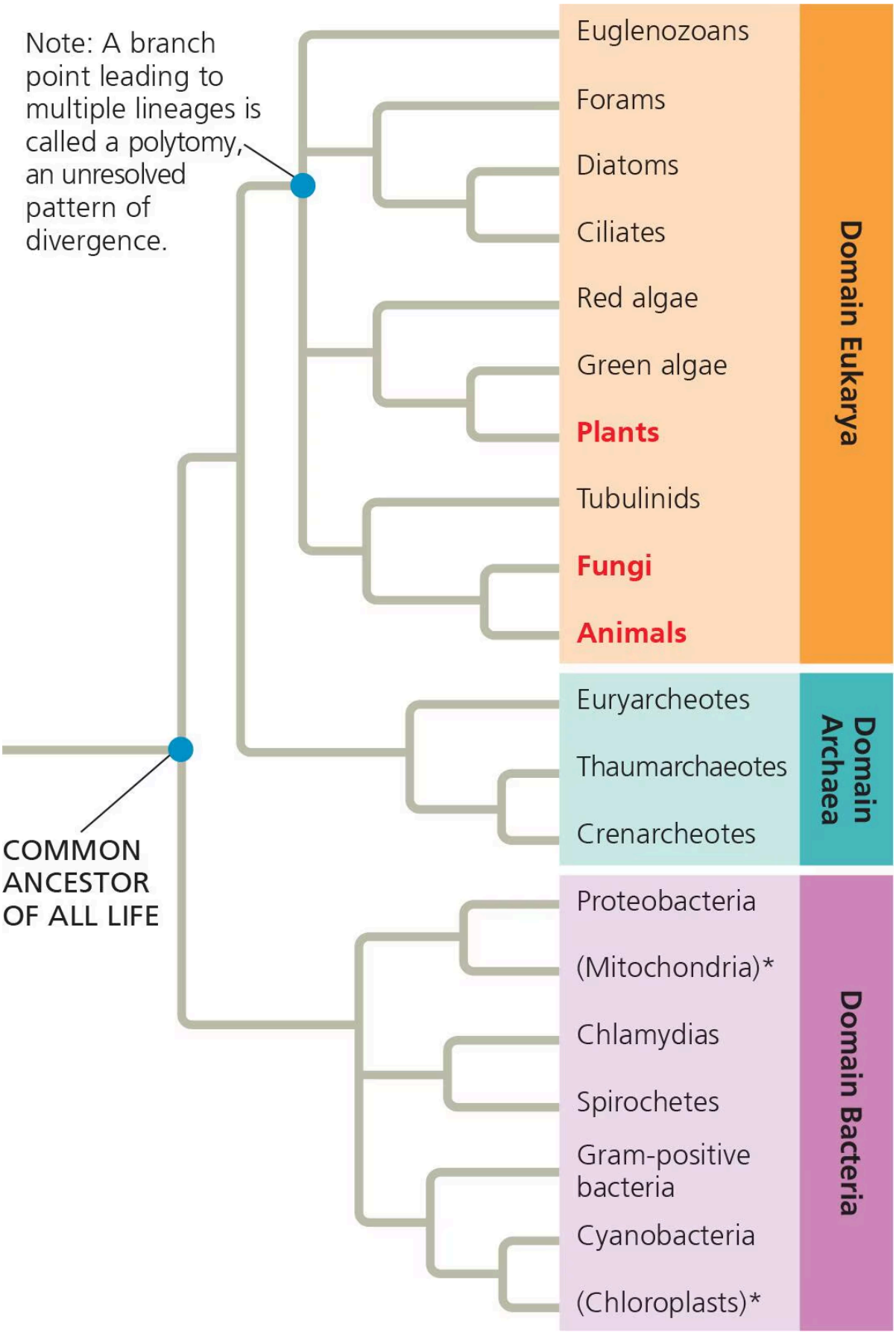
Note: A branch point leading to multiple lineages is called a polytomy, an unresolved pattern of divergence.
COMMON ANCESTOR OF ALL LIFE
Euglenozoans
Forams
Diatoms
Ciliates
Red algae
Green algae
Plants
Tubulinids
Fungi
Animals
Domain Eukarya
Euryarcheotes
Thaumarchaeotes
Crenarcheotes
Domain Archaea
Proteobacteria
(Mitochondria)*
Chlamydias
Spirochetes
Gram-positive bacteria
Cyanobacteria
(Chloroplasts)*
Domain Bacteria

This phylogenetic tree is based on sequence data for rRNA and other genes. For simplicity, only some of the major branches in each domain are shown. Lineages within Eukarya that are dominated by multicellular organisms (plants, fungi, and animals) are in red type, while the two lineages denoted by an asterisk are based on DNA from cellular organelles. All other lineages consist solely or mainly of single-celled organisms.

The tree is as follows. Root leads to node 1, common ancestor of all life. Node 1 splits to nodes 2 and 3. Node 2 splits to nodes 4 and 5. Node 4 splits to branch tip euglenozoans and nodes 6, 7, and 8. Node 6 splits to branch tip forams and node 9. Node 9 splits to branch tips diatoms and ciliates. Node 7 splits to branch tip red algae and node 10. Node 10 splits to branch tips green algae and plants. Node 8 splits to branch tip tubulinds and node 11. Node 11 splits to branch tips fungi and animals. Node 5 splits to branch tip euryarcheotes and node 12. Node 12 splits to branch tips thaumarchaeotes and crenarcheotes. Node 3 splits to nodes 13, 14, and 15. Node 13 splits to branch tips proteobacteria and mitochondria. Node 14 splits to branch tips chlamydias and spirochetes. Node 15 splits to branch tip gram-positive bacteria and node 16. Nod 16 splits to branch tips cyanobacteria and chloroplasts. Branches euglenozoans through animals represent domain Eukarya. Branches euryarcheotes through crenarcheotes represent domain Archaea. Branches proteobacteria through chloroplasts represent domain bacteria. Note: A branch point leading to multiple lineages is called a polytomy, an unresolved pattern of divergence. In this figure, one example of polytomy is the common point from which spring the various branches of the domain Eukarya.

MAKE CONNECTIONS *After reviewing endosymbiont theory (see*Figure 6.16*)*, explain the specific positions of the mitochondrion and chloroplast lineages on this tree.*

The three-domain system highlights the fact that much of the history of life has been about single-celled organisms. The two prokaryotic domains consist entirely of single-celled organisms, and even in Eukarya, only the branches labeled in red type (plants, fungi, and animals) are dominated by multicellular organisms. Of the five kingdoms previously recognized by systematists, most biologists continue to recognize Plantae, Fungi, and Animalia, but not Monera and Protista. The kingdom Monera is obsolete because it would have members in two different domains. The kingdom Protista has also crumbled because it includes members that are more closely related to plants, fungi, or animals than to other protists (see Figure 28.5).

New research continues to change our understanding of the tree of life. For example, recent metagenomic studies have uncovered the genomes of many new species of archaea, leading to discovery of the Thaumarchaeota and other previously unknown phyla of archaea. One of these newly discovered groups, the lokiarcheaota, may represent the long-sought-after sister group of the eukaryotes. If confirmed, this discovery could help shed light on how eukaryotes arose from their prokaryotic ancestors (see Concepts 27.4 and 28.1).

The Important Role of Horizontal Gene Transfer

In the phylogeny shown in Figure 26.21, the first major split in the history of life occurred when bacteria diverged from other organisms. If this tree is correct, eukaryotes and archaea are more closely related to each other than either is to bacteria.

This reconstruction of the tree of life is based in part on sequence comparisons of rRNA genes, which code for the RNA components of ribosomes. However, some other genes reveal a different set of relationships. For example, researchers have found that many of the genes that influence metabolism in yeast (a unicellular eukaryote) are more similar to genes in the domain Bacteria than they are to genes in the domain Archaea—a finding that suggests that the eukaryotes may share a more recent common ancestor with bacteria than with archaea.

What causes trees based on data from different genes to yield such different results? Comparisons of complete genomes from the three domains show that there have been substantial movements of genes between organisms in the different domains. These took place through **horizontal gene transfer**, a process in which genes are transferred from one genome to another through mechanisms such as exchange of transposable elements and plasmids, viral infection (see Concept 19.2), and fusions of organisms (as when a host and its endosymbiont become a single organism). Recent research reinforces the view that horizontal gene transfer is important. For example, one study found that on average, 80% of the genes in 181 prokaryotic genomes had moved between species at some point during the course of evolution. Because phylogenetic trees are based on the assumption that genes are passed vertically from one generation to the next, the occurrence of such horizontal transfer events helps to explain why trees built using different genes can give inconsistent results.

Horizontal gene transfer can also occur between eukaryotes. For example, over 200 cases of the horizontal transfer of transposons have been reported in eukaryotes, including humans and other primates, plants, birds, and lizards. Nuclear genes have also been transferred horizontally from one eukaryote to another. The **Scientific Skills Exercise** describes one such example, giving you the opportunity to interpret data on the transfer of a pigment gene to an aphid from another species.

Scientific Skills Exercise

Using Protein Sequence Data to Test an Evolutionary Hypothesis

AP Science Practices 2, 4, 5, 6

Did Aphids Acquire Their Ability to Make Carotenoids Through Horizontal Gene Transfer?
Carotenoids are colored molecules that have diverse functions in many organisms, such as photosynthesis in plants and light detection in animals. Plants and many microorganisms can synthesize carotenoids from scratch, but animals generally cannot (they must obtain carotenoids from their diet). One exception is the pea aphid *Acyrthosiphon pisum*, a small, plant-dwelling insect whose genome includes a full set of genes for the enzymes needed to make carotenoids. Because other animals lack these genes, it is unlikely that aphids inherited them from a single-celled common ancestor shared with microorganisms and plants. So where did they come from? Evolutionary biologists hypothesized that an aphid ancestor acquired these genes by horizontal gene transfer from distantly related organisms.

How the Experiment Was Done Scientists obtained the DNA sequences for the carotenoid-biosynthesis genes from several species, including aphids, fungi, bacteria, and plants. A computer "translated" these sequences into amino acid sequences of the encoded polypeptides and aligned the amino acid sequences. This allowed the team to compare the corresponding polypeptides in the different organisms.

Data from the Experiment The sequences below show the first 60 amino acids of one polypeptide of the carotenoid-biosynthesis enzymes in the plant *Arabidopsis thaliana* (bottom) and the corresponding amino acids in five nonplant species, using the one-letter abbreviations for the amino acids (see Figure 5.14). A dash – indicates a gap inserted in a sequence to optimize its alignment with the corresponding sequence in *Arabidopsis*.

Organism	Alignment of Amino Acid Sequences					
Acyrthosiphon (aphid)	IKIIIIGSGV	GGTAAAARLS	KKGFQVEVYE	KNSYNGGRCS	IIR-HNGHRF	DQGPSL––YL
Ustilago (fungus)	KKVVIIGAGA	GGTALAARLG	RRGYSVTVLE	KNSFGGGRCS	LIH-HDGHRW	DQGPSL––YL
Gibberella (fungus)	KSVIVIGAGV	GGVSTAARLA	KAGFKVTILE	KNDFTGGRCS	LIH-NDGHRF	DQGPSL––LL
Staphylococcus (bacterium)	MKIAVIGAGV	TGLAAAARIA	SQGHEVTIFE	KNNNVGGRMN	QLK-KDGFTF	DMGPTI––VM
Pantoea (bacterium)	KRTFVIGAGF	GGLALAIRLQ	AAGIATTVLE	QHDKPGGRAY	VWQ-DQGFTF	DAGPTV––IT
Arabidopsis (plant)	WDAVVIGGGH	NGLTAAAYLA	RGGLSVAVLE	RRHVIGGAAV	TEEIVPGFKF	SRCSYLQGLL

Data from Nancy A. Moran, Yale University. See N. A. Moran and T. Jarvik, Lateral transfer of genes from fungi underlies carotenoid production in aphids, *Science* 328:624–627 (2010).

Organism	Alignment of amino acid sequences
Acyrthosiphon (aphid)	I K I I I I I G S G Y
Ustilago (fungus)	K K V V I I G A G A
Glbberella (fungus)	K S V I V I G A G V
Staphylococcus (bacterium)	M K I A V I G A G V
Pantoea (bacterium)	K R T F V I G A G F
Arabidopsis (plant)	W D A V V I G G G H

INTERPRET THE DATA

- In the rows of data for the organisms being compared with the aphid, highlight the amino acids that are identical to the corresponding amino acids in the aphid.
- Which organism has the most amino acids in common with the aphid? Rank the partial polypeptides from the other four organisms in degree of similarity to that of the aphid.
-
    - Do these data support the hypothesis that aphids acquired the gene for this polypeptide by horizontal gene transfer? Why or why not?

- If horizontal gene transfer did occur, what type of organism is likely to have been the source?
- What additional sequence data would support your hypothesis?
- How would you account for the similarities between the aphid sequence and the sequences for the bacteria and plant?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Studies show that eukaryotes can even acquire genes by horizontal gene transfer from prokaryotes. For example, a recent genomic analysis showed that the alga *Galdieria sulphuraria***(Figure 26.22)** acquired about 5% of its nuclear genes from various bacterial and archaeal species. Unlike most eukaryotes, this alga can survive in environments that are highly acidic or extremely hot, as well as those with high concentrations of heavy metals. The researchers identified specific genes transferred from prokaryotes that have enabled *G. sulphuraria* to thrive in such extreme habitats.

Figure 26.22

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A recipient of transferred genes: the alga *Galdieria sulphuraria.*
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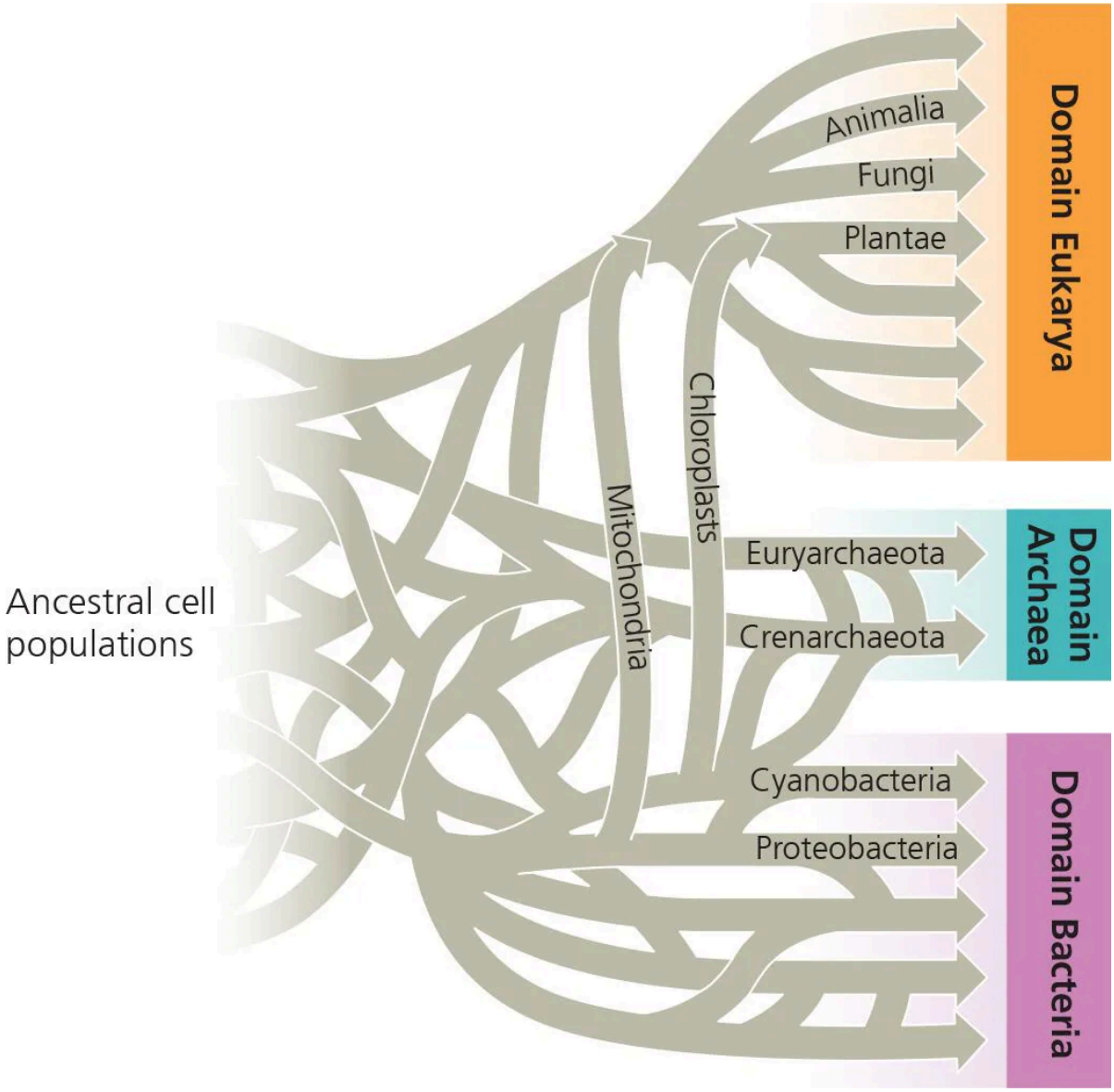
Genes received from prokaryotes enable G. sulphuraria (inset) to grow in extreme environments, including on sulfur-encrusted rocks around volcanic hot springs similar to this one in Yellowstone National Park.

Overall, horizontal gene transfer has played a key role throughout the evolutionary history of life, and it continues to occur today. Some biologists have argued that horizontal gene transfer was so common that the early history of life should be represented not as a dichotomously branching tree like that in Figure 26.21, but rather as a tangled network of connected branches **(Figure 26.23)**. Although scientists continue to debate how best to portray the earliest steps in the history of life, in recent decades there have been many exciting discoveries about evolutionary events that occurred over time. We'll explore such discoveries in the rest of this unit, beginning with Earth's earliest inhabitants, the prokaryotes.

Figure 26.23

A tangled web of life.



Horizontal gene transfer may have been so common in the early history of life that the base of a "tree of life" might be more accurately portrayed as a tangled web.

Tangled web diagram for tree of life, starting at far left with ancestral cell populations, moving through large tangled web to domains eukarya, archaea, and bacteria on the right. Domain Eukarya has 7 branches, including Animalia, Fungi, and Plantae (others are unlabeled). Domain Archaea, directly below Domain Eukarya, has two branches, the Euryarchaeota and the Crenarchaeota. Webbing connects these two. Domain Bacteria, below Domain Archaea, has five branches, including the Cyanobacteria and the Proteobacteria. Others are unlabeled, and webbing connects some of these branches. Approximately halfway between the ancestral cell populations (far left of the diagram) and the three domains of life (far right of the diagram), webbing labeled chloroplasts and mitochondria links the cyanobacteria (chloroplasts) and the proteobacteria (mitochondria) to the plantae (chloroplasts) and to the common ancestor of all eukaryotes (mitochondria). This webbing is shown as an arrow indicating direction of flow is from Domain Bacteria to Domain Eukarya.

Interview

Interview with Nancy Moran: Studying symbioses and horizontal gene transfer